

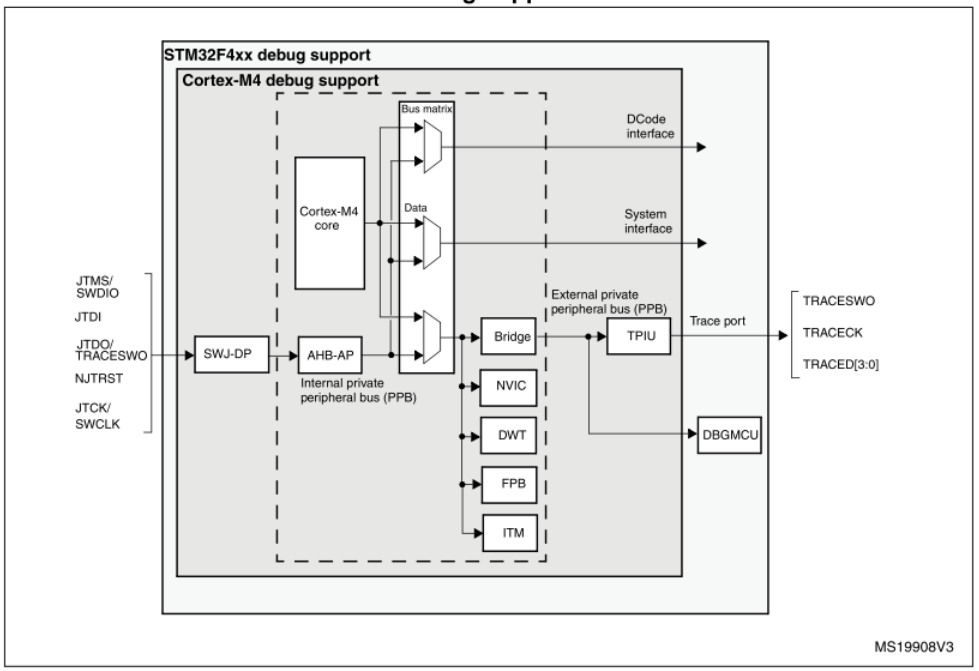
Using SWO to print information from STM32

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Using SWO

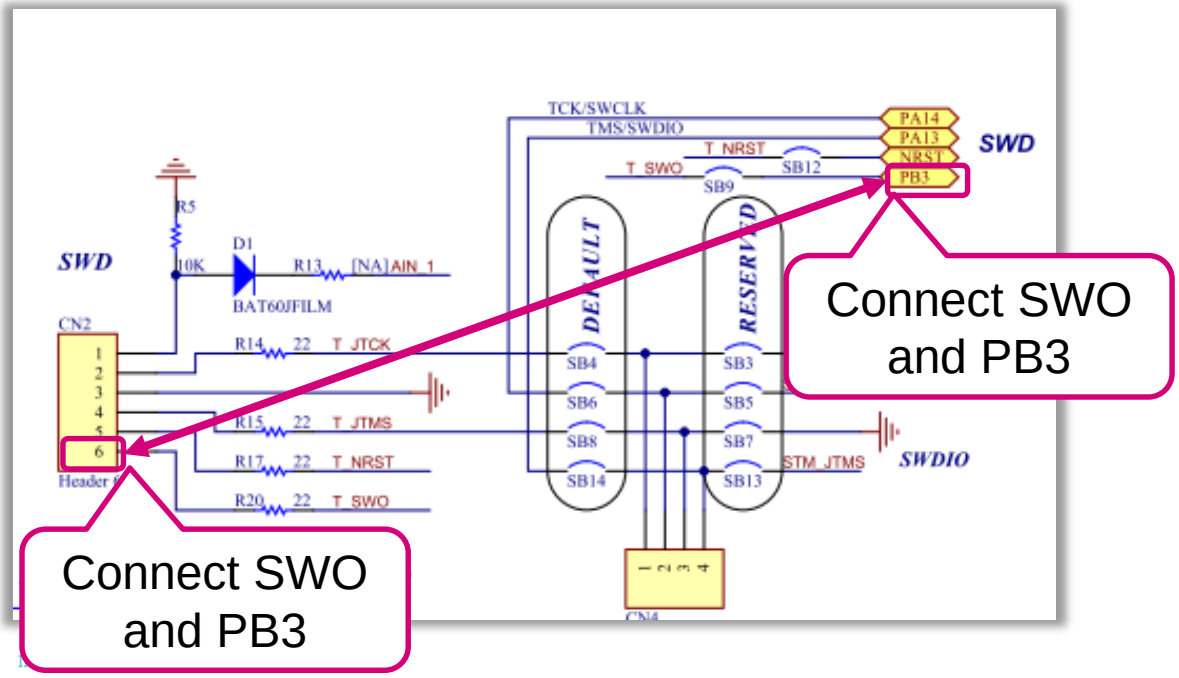
- On some stm32 is periphery called ITM, not mix with ETM(real trace)
- This periphery can be used to internal send data from MCU over SWO pin
- Is possible to redirect the printf into this periphery
- And also some IDEs can display this information during debug
- It is similar to USART but we don’t need any additional wires and PC terminal
- Video: Link

Figure 483. Block diagram of STM32 MCU and Cortex®-M4 with FPU-level debug support



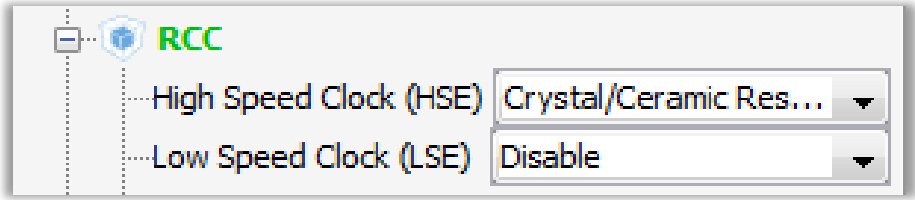
Using SWO

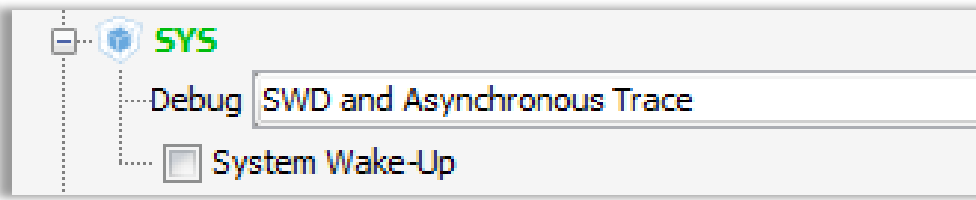
- To make SWO working you musty connect the PB3 and Debugger SWO pin together

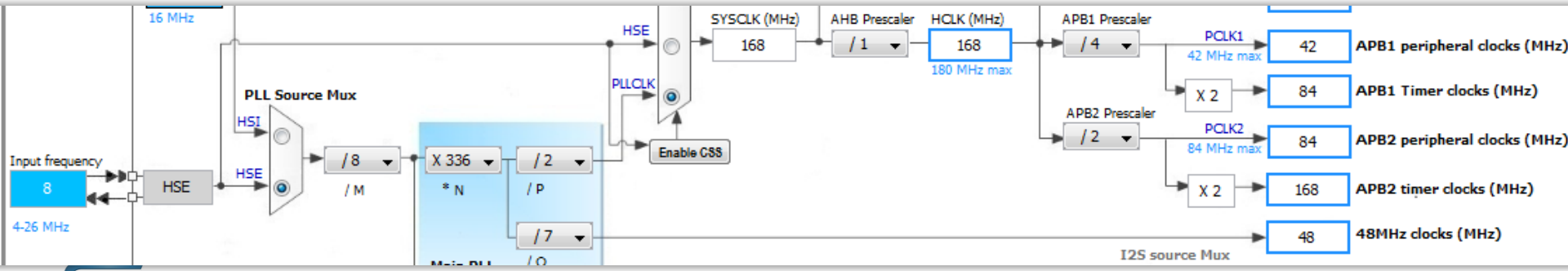


Using SWO for printf

- Create project in CubeMX
  - Menu > File > New Project
  - Select STM32F4 > STM32F429/439 > LQFP144 > STM32F439ZITx
- We need only blank project with clock initialization
  - We set the RCC and configure the core to maximum speed and SWD with SWO

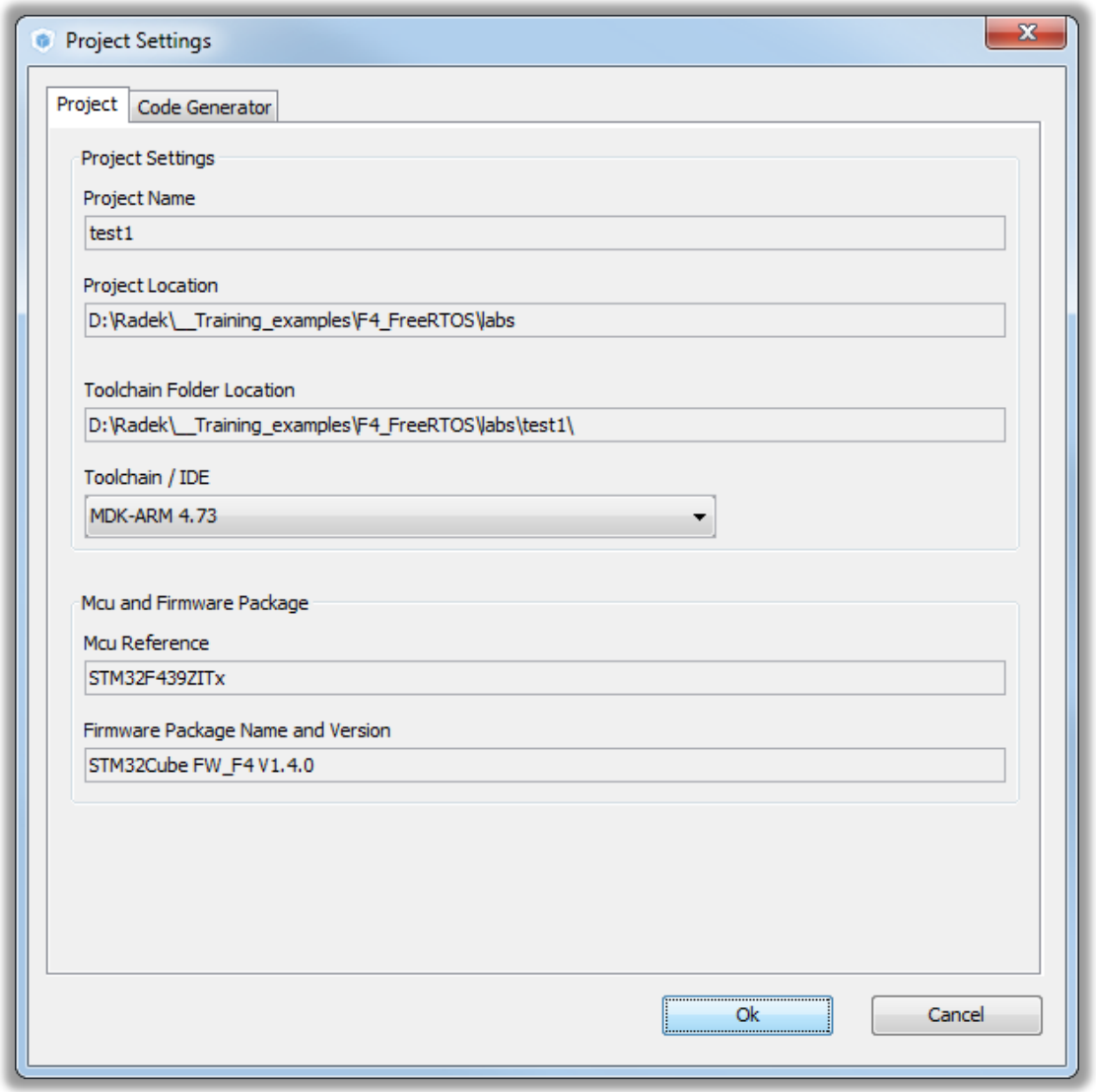






Using SWO for printf

- Now we set the project details for generation
  - Menu > Project > Project Settings
  - Set the project name
  - Project location
  - Type of toolchain
- Now we can Generate Code
  - Menu > Project > Generate Code



Using SWO for printf in KEIL

- We need to include the stdio.h library to make printf working

```
/* USER CODE BEGIN Includes */
#include <stdio.h>
/* USER CODE END Includes */
```

- And define __FILE structure

```
/* USER CODE BEGIN PFP */
struct __FILE { int handle; /* Add whatever is needed */ };
/* USER CODE END PFP */
```

- fputc function must be defined to send byte over ITM

```
/* USER CODE BEGIN 4 */
/*send text over SWV*/
int fputc(int ch, FILE *f) {
ITM_SendChar(ch);//send method for SWV
  return(ch);
}
/* USER CODE END 4 */
```

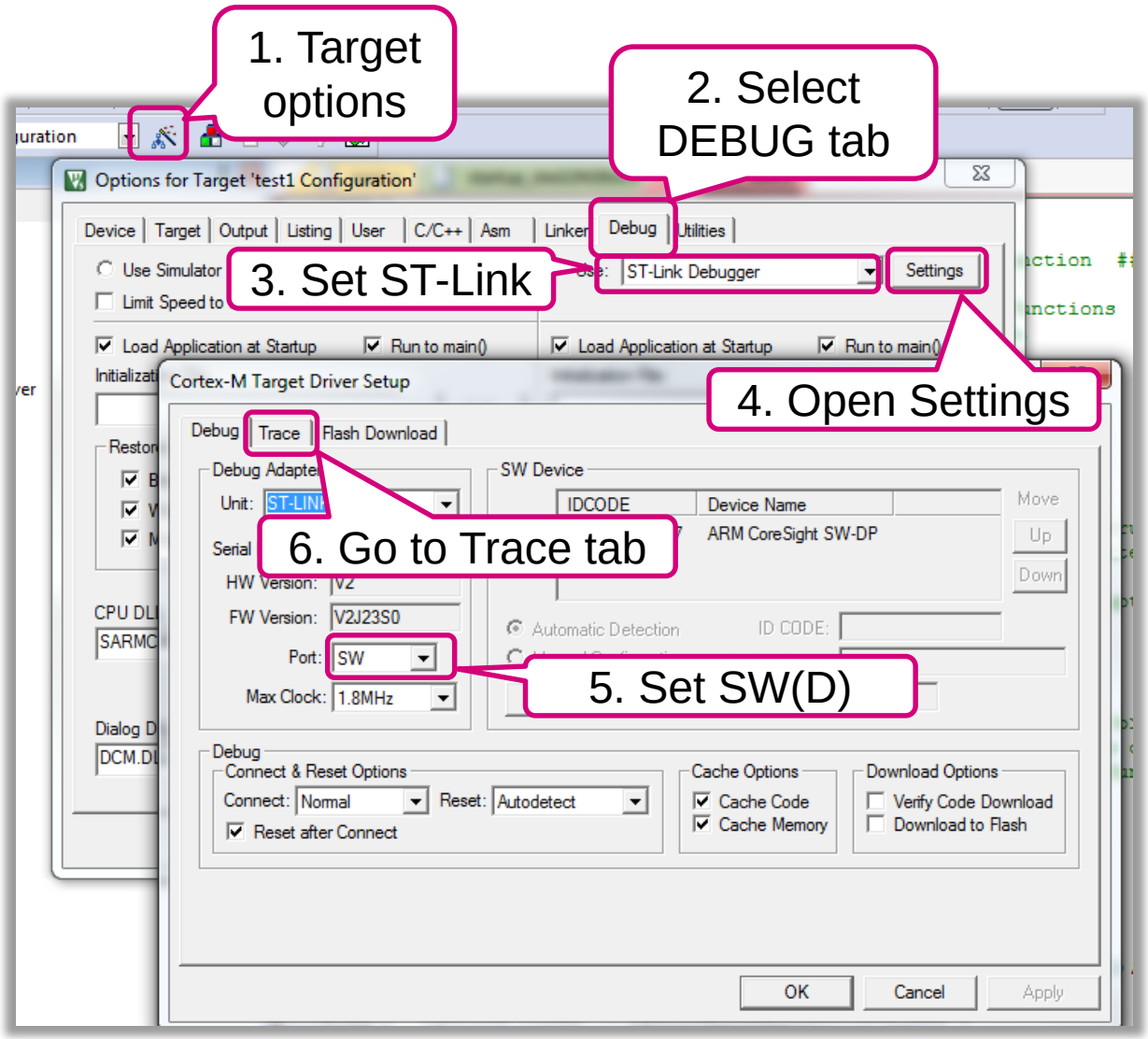
Using SWO for printf in KEIL

- If the MCU run on very high frequency and you not see print f output you may try put into ITM send delay loop

```
/* USER CODE BEGIN 4 */
/*send text over SWV*/
int fputc(int ch, FILE *f) {
  uint32_t i=0;
  for(i=0;i<0xFFFF;i++);//waiting method, lower value will stop the SWV
  ITM_SendChar(ch);//send method for SWV
  return(ch);
}
/* USER CODE END 4 */
```

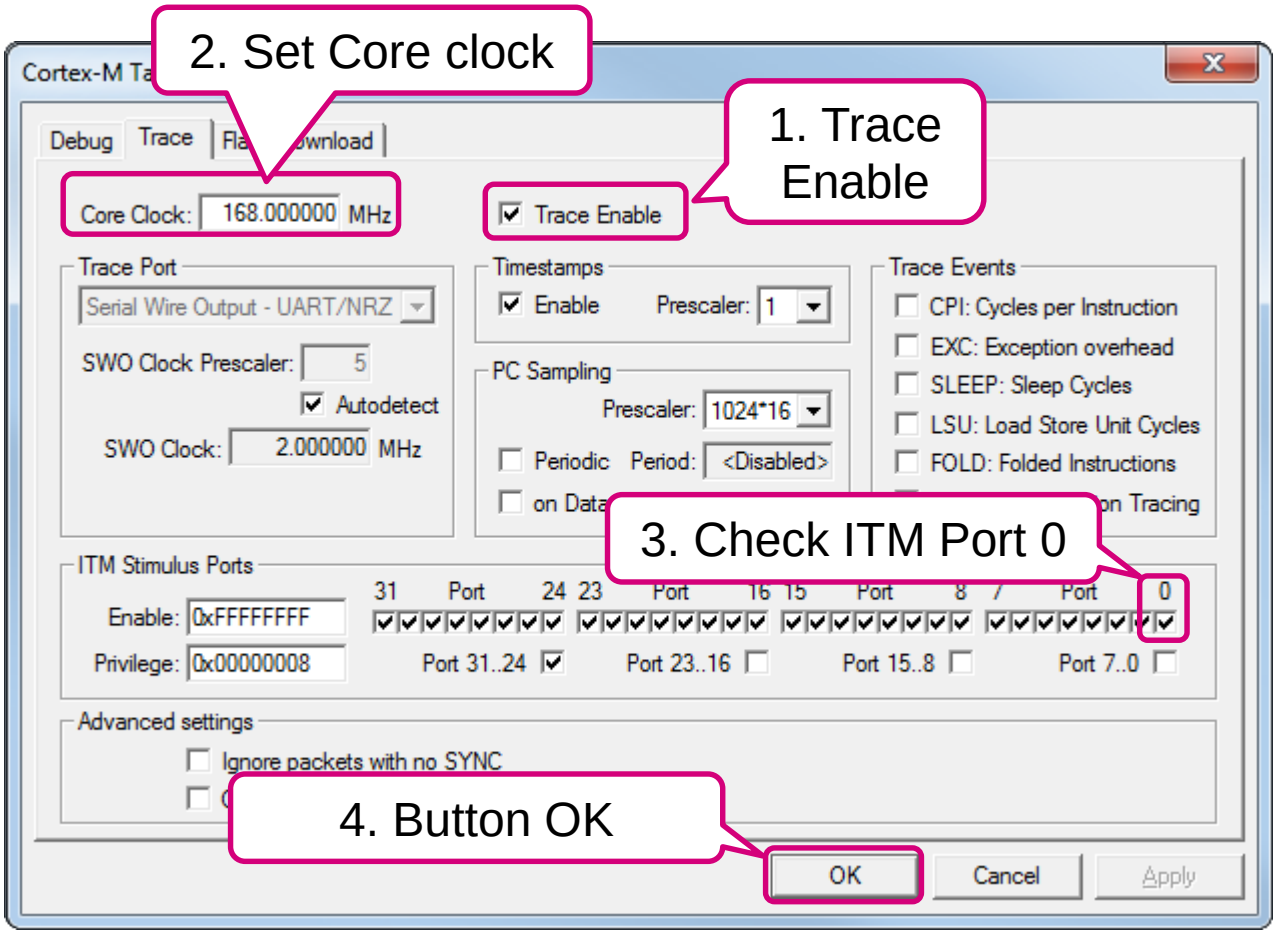
Using SWO for printf in KEIL

- In KEIL
- Open target options (ALT+F7)
- Set ST-Link debugger
- Button Settings
- Set SW(D) in Debug TAB
- Got to Trace TAB



Using SWO for printf in KEIL

- Check Trace Enable
- Set core clock (168MHz must be same as in CubeMX clock tree)
- ITM Stimulus Port 0 must be checked
- Button OK



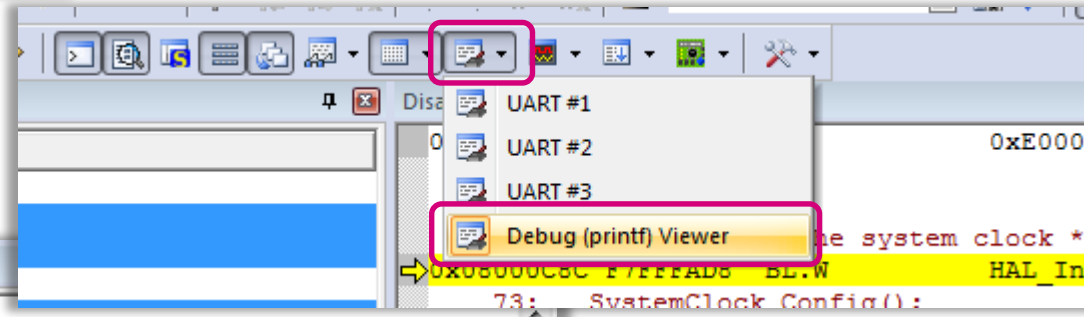
Using SWO for printf in KEIL

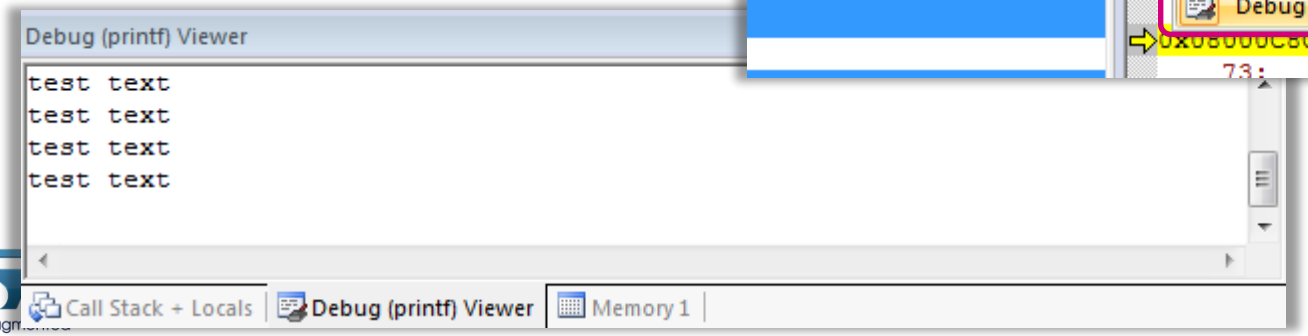
- Testing loop printf in main

```
/* USER CODE BEGIN 3 */
/* Infinite loop */
while (1)
{
  printf("test text\n");
  HAL_Delay(1000);
}
/* USER CODE END 3 */
```

- Compile and debug project

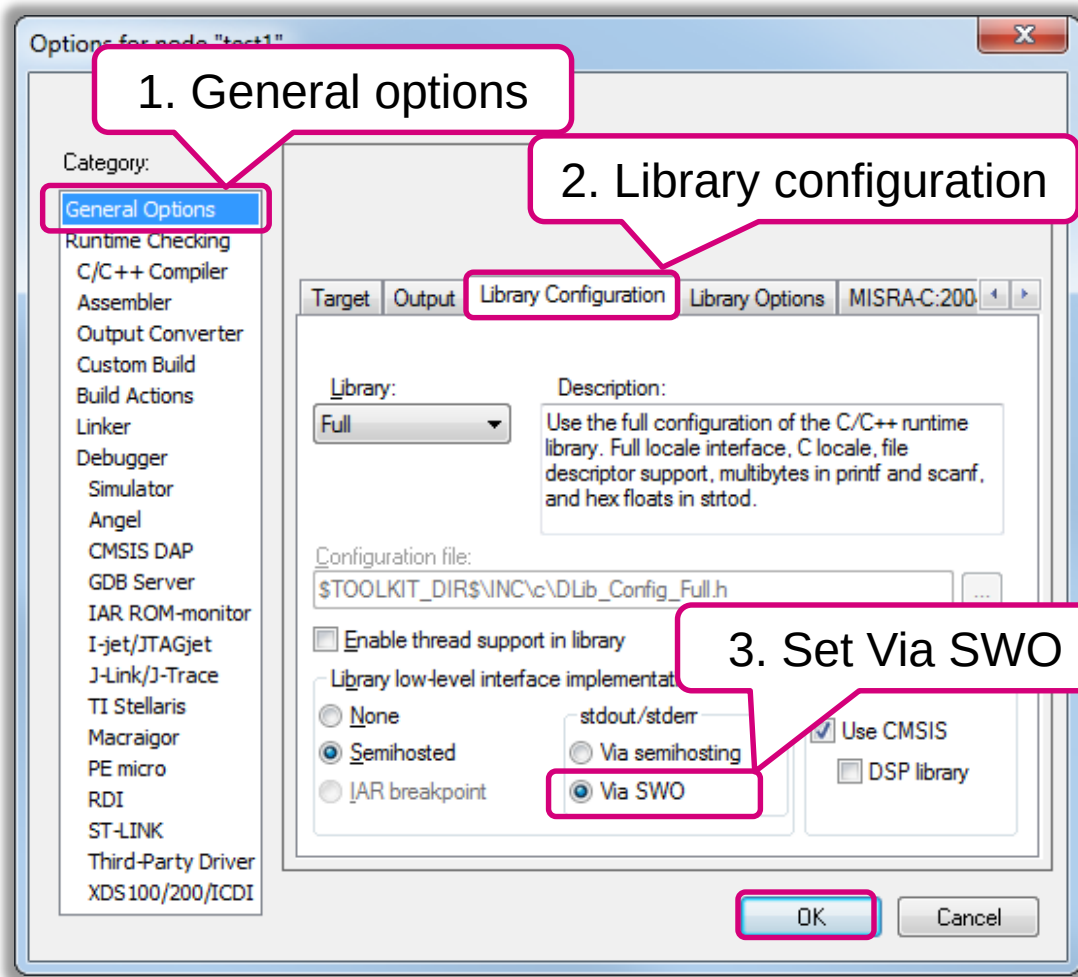
- Open Debug (printf) Viewer





Using SWO for printf in IAR

- Here is configuration more easier
- Open project options
- General Options
- TAB library and configuration
- Select stdout/stderr via SWO
- Button OK



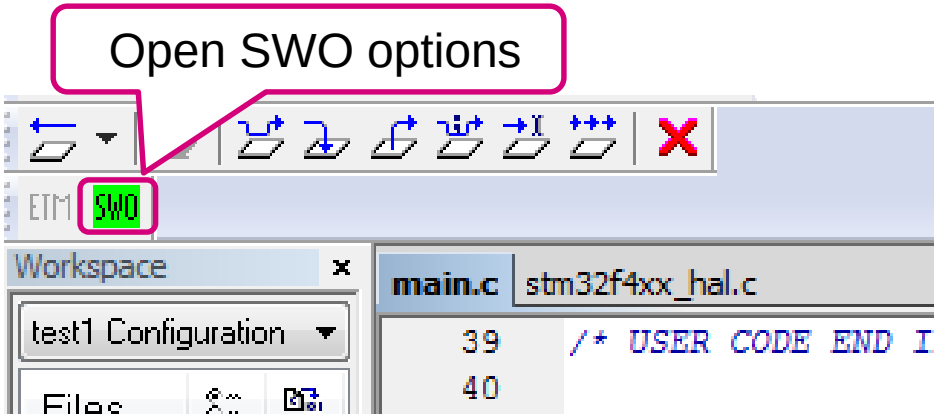
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Using SWO for printf in IAR

- Add testing printf into loop

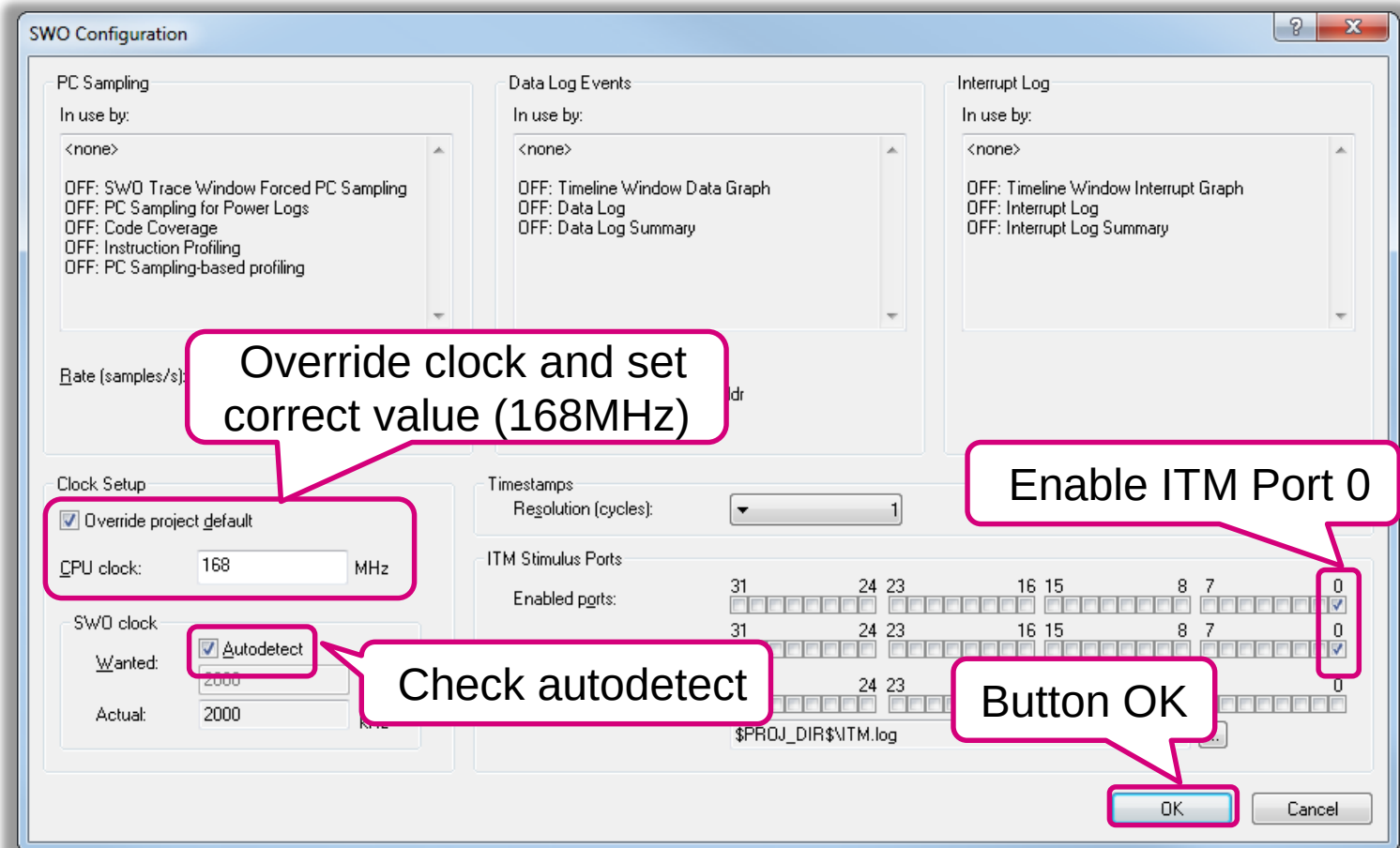
```
/* USER CODE BEGIN 3 */
/* Infinite loop */
while (1)
{
  printf("test text\n");
  HAL_Delay(1000);
}
/* USER CODE END 3 */
```

- Compile project and go to debug
- Open SWO



Using SWO for printf in IAR

- Override project default if clocks are different from core (168MHz in out case)
- Check autodetect SWO clock if possible
- Enable ITM Port 0 to terminal
- Button OK



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Using SWO for printf in IAR

- Menu View -> Terminal I/O
- Run program



